

CHAPTER 3

Conceptual Foundations of Functional Behavioral Assessment

Those who fall in love with practice without science are
like a sailor who sails a ship without a helm or compass,
and can never be certain whither he is going. . . .

—LEONARDO DA VINCI (1452–1519)

Applied behavior analysis is the science of human behavior. FBA is the process of applying these scientific principles to assess human behavior and to develop interventions that lead to socially meaningful behavioral outcomes. Before discussing FBA procedures, we think it is essential to establish a common understanding of the basic terminology, principles, and concepts of applied behavior analysis. Having said that, we are also assuming that our readers have an understanding of the basic principles of learning. The goal of this chapter is not to *teach* foundational principles and concepts related to positive and negative reinforcement, positive and negative punishment, stimulus control, discriminative stimuli, motivating operations, matching law, and so forth. Rather, the goal is to provide a *review* or refresher of these principles and concepts. Our emphasis is on identifying the variables that evoke, occasion, and reinforce interfering behaviors and then synthesizing the results of FBAs to arrive at a conceptual understanding of student behavior. Later, we will refer to this process as a *conceptual synthesis*.

FUNCTIONAL ANTECEDENTS AND CONSEQUENCES

When conducting an FBA it is critical to differentiate between relevant and irrelevant stimuli that influence interfering behavior. In other words, it is important to distinguish between stimuli that are *associated* with interfering behavior and those that are *functionally related* to interfering behavior. When conducting an FBA, we are concerned with identifying and describing the variables that (1) make interfering behavior more likely (functional antecedents) and (2) strengthen and maintain behavior (functional consequences/reinforcers).

Separating the Wheat from the Chaff

The purpose of FBA is to:

- Weed out associated and irrelevant variables, and
- Identify and describe the variables that are functionally related to the interfering behavior.

Many stimuli occur prior to occurrences of interfering behavior. Some of these stimuli are functionally related to the interfering behavior, while others are not. Imagine, for example, that a school psychologist conducted an anecdotal observation of a student with a history of interfering behaviors (e.g., swearing and arguing). At the start of this observation, the school psychologist noticed that a classmate began humming a tune immediately before the referred student engaged in interfering behaviors. Does this mean that the classmate's humming a tune is a functional antecedent that triggered the student's interfering behaviors? Perhaps, but not likely . . .

After conducting interviews, the school psychologist learned that just prior to the observation, the classroom teacher had delivered the third verbal and gestural cue for the student to disengage from socializing with a peer and begin the assignment. Subsequent observations further suggested that repeated cues from the teacher typically preceded occurrences of interfering behavior. In this case, the classmate's humming was a *neutral* or *irrelevant* stimulus, and the redirection cues delivered by the teacher were *functional antecedents*.

CONSEQUENCE VARIABLES

When teaching the conceptual foundations of FBA, we have found that reviewing the principles of reinforcement and punishment facilitates an understanding of antecedent influences on behavior, so we begin with a review of behavioral consequences. *Consequences* refer to the effects or outcomes of behavior. Functional consequences are the events that occur after a behavior and alter the future likelihood of similar behaviors. Consequences that follow and strengthen behaviors (i.e., increase the future frequency, duration, and/or intensity of behaviors) are called *reinforcers*. The process by which behavior is strengthened when it is followed by a consequence is called *reinforcement*. In contrast, consequences that follow and weaken behaviors (i.e., decrease the future frequency, duration, and/or intensity of behaviors) are called *punishers*. And the process by which a behavior is weakened when it is followed by a consequence is called *punishment*. When conducting an FBA, we are primarily interested in the reinforcing consequences that strengthen and maintain interfering behaviors. Therefore, we focus our discussion of behavioral consequences on the two types of reinforcing consequences: positive reinforcement and negative reinforcement.

Positive Reinforcement

Behaviors that produce desirable outcomes are likely to be repeated. When a behavior results in access to preferred stimuli or events (e.g., social attention from a close friend, a favorite food or activity, monetary rewards) and therefore becomes more likely to occur in the future, we refer to the consequences as *positive reinforcers* and to the process as *positive reinforcement*. Humans, by their very nature, are social beings. Thus, attention from others is often a powerful positive reinforcer for behavior. Let's take a look at a familiar example to illustrate the power of social attention as a positive reinforcer.

Example of Positive Reinforcement (Social Attention)

Kent is a seventh-grade student with average-to-above-average cognitive abilities. According to his teachers, Kent's daily classroom performance and grades do not reflect his true potential. In fact, they have reported that Kent's classroom behavior is quite disruptive and annoying. When asked to describe what Kent does that is so annoying, his teachers explained that he often makes wisecracks during independent seatwork periods. They also acknowledged that his comments are rather funny and tend to evoke at least mild chuckles from classmates. Direct observations of Kent's behavior confirmed the teachers' reports about his wisecracking behavior, the inherent humor in what he says, and—perhaps most importantly—the response of his classmates: laughter or groans. In this example, the response from Kent's classmates (i.e., laughter or groans) *positively reinforced* his wisecrack remarks. Thus, his wisecracking comments are likely to continue for as long as his peers respond in this manner.

Negative Reinforcement

Behaviors that successfully result in avoidance or termination of aversive circumstances also tend to be repeated. Everyone considers some activities to be enjoyable and others to be difficult, uncomfortable, or otherwise aversive. And we tend to shy away from items and activities that are sufficiently unpleasant and have only minimal reinforcing value. Most adults, for example, do not choose to engage in behaviors that they do not find fun in some way unless the payoff is personally significant (consider exercising!). *Negative reinforcement* occurs when an interfering behavior results in the postponement (avoidance) or termination (escape) of an aversive circumstance. Below is an example of how negative reinforcement may operate in the classroom.

Example of Negative Reinforcement (Escape from Task)

Creed is an 8-year-old who exhibits interfering behaviors such as throwing materials, yelling at the teacher, climbing on desks, and running out of the classroom. His teacher, Ms. Timmerman, observed that his outbursts seemingly occurred randomly throughout the school day. She tried every trick in her disciplinary bag, from a “corner time-out” to keeping him in the classroom during recess. As the number of incidents began to pile up, she resorted to sending Creed to the principal's office, where he was directed to sit in a chair for 15–20 minutes. Ms. Timmerman, in a somewhat bewildered tone, noted that Creed's behavior had not gotten any better. After conducting an FBA, the school practitioner recognized that Creed's outbursts did indeed follow a pattern. In fact, almost all of his outbursts occurred when writing was required to complete an assignment. Creed used a pencil to fill in ovals on tests without incident, but each time he was required to produce actual written work, regardless of the subject matter, an outburst occurred. Thus, the school practitioner identified assignments that required writing as functional antecedents and escape from tasks (e.g., being sent to time-out) as the functional consequence. Essentially, Creed had learned through numerous interactions in the classroom that disruptive outbursts allowed him to avoid or escape assignments that involved writing, and the teacher's disciplinary consequences *negatively reinforced* interfering behaviors.

There's More Than One Way to Treat a Function

In their comprehensive review of the behavior-analytic research, Geiger, Carr, and LeBlanc (2010) identified and described the following eight interventions to consider when developing function-focused interventions for interfering behaviors maintained by negative reinforcement in the form of escape from “demands” or “instructional opportunities.”

1. *Activity choice*. Offer students the opportunity to choose the order in which assigned tasks are completed.
2. *Curricular and instructional revisions*. Modify the task expectations or instructional procedures to make assigned activities less aversive.
3. *Demand fading*. Remove all nonpreferred instructional opportunities (demands) and then gradually reintroduce them while avoiding or minimizing negative reinforcement of interfering behaviors.
4. *Differential negative reinforcement of alternative behavior*. Provide a brief period of “escape” in the form of a break from work contingent on a desired replacement behavior (e.g., accurate work completion).
5. *Differential negative reinforcement of zero rates of responding*. Provide a brief period of “escape” in the form of a break from work when a specified period of time elapses without occurrences of the interfering behavior.
6. *Extinction*. Do not permit escape from work contingent on interfering behavior; continue presenting the task regardless of the interfering behavior.
7. *Functional communication training*. Teach and reinforce an appropriate communicative response (e.g., requests for a break) and place the interfering behavior on extinction.
8. *Noncontingent escape*. Provide brief breaks from work on a time-based schedule, irrespective of interfering behavior.

Check out Chapter 11 for additional function-focused treatment strategies!

Individually Mediated, Socially Mediated, and Automatic Reinforcement

Reinforcement may be individually mediated, socially mediated, or automatic. With *individually mediated reinforcement*, the behavior produces the reinforcement directly and independent of others. With *socially mediated reinforcement*, someone else delivers reinforcement following the behavior. And with *automatic reinforcement*, the behavior directly and immediately produces a biological or sensory consequence. Consider the following examples to illustrate the distinctions between each form of reinforcement.

Individually Mediated Reinforcement

- You are thirsty, so you walk to the kitchen and open the refrigerator. When you retrieve a preferred icy-cold beverage, you experience *individually mediated positive reinforcement*. The behavior of opening the refrigerator and reaching for a beverage produced direct access to a preferred item.
- You are in a casino operating a slot machine. The behavior of pulling the lever of the “one-armed bandit” (or, nowadays, pushing a button on the slot machine) eventually produces a payload of coins (or, nowadays, bells, whistles, and a paper receipt). The behavior of operating the slot machine therefore resulted in *individually mediated positive reinforcement*.

- You need to begin the laborious process of writing a psychological evaluation report but decide to check your e-mail instead. In this case, checking your e-mail represents a set of behaviors that produce both *individually mediated positive reinforcement* (access to valuable messages from others) and *individually mediated negative reinforcement* (avoidance of report writing). The action of checking e-mail resulted in these consequences directly, and you were all by yourself!

Socially Mediated Reinforcement

- A student solves a complicated math problem independently, and the teacher delivers enthusiastic verbal praise to increase the likelihood of independent work completion in the future. In this case, the teacher delivered *socially mediated positive reinforcement*.
- A paraprofessional verbally directs a student with a diagnosis of social phobia to leave the classroom setting and go to the “cool-down room” when she engages in interfering behaviors. The student engages in interfering behaviors when expected to participate in small-group learning activities, which she considers highly aversive. In this case, the student’s interfering behaviors resulted in *socially mediated negative reinforcement*.

Automatic Reinforcement

- A young girl slowly and gently twirls her hair when she is alone in her room. The hair-twirling behavior produces pleasant sensations (e.g., the feeling of the hair twisting in her hands and the slight tugging at her scalp). In this example, hair-twirling behavior produced *automatic positive reinforcement*.
- A student engages in hand-flapping behavior that results in a pleasant form of visual stimulation. In this example, hand-flapping behavior produced *automatic positive reinforcement*.
- You have several mosquito bites, and the intense itching is “driving you crazy.” When you scratch those little lumps, it temporarily reduces the intensity of the itching. In this case, scratching behavior produced *automatic negative reinforcement*.
- Another young girl twirls her hair when alone. In this case, however, hair twirling occurs with a high level of intensity and results in the loss of hair (trichotillomania). When interviewed, she reports that she typically twirls her hair when she feels tense and that the behavior makes her feel better. In this case, hair-twirling behavior resulted in *automatic negative reinforcement* by alleviating/reducing an aversive internal state.

These examples illustrate three key ideas. First, both individually and socially mediated reinforcement involve consequences that are directly observable, whereas automatic reinforcement results in an unobservable (private) event or physiological sensation. Second, individually mediated, socially mediated, and automatic reinforcement may be further classified as positive or negative (see Table 3.1). When the behavior results in access to preferred social interactions, items, activities, or sensory stimulation, we classify those consequences as positive reinforcers. In contrast, when the behavior results in a delay, reduc-

TABLE 3.1. Classification of Reinforcement Contingencies

Reinforcement	Individually mediated	Socially mediated	Automatic
Positive (+)	Individually mediated positive reinforcement (direct access)	Socially mediated positive reinforcement (socially mediated access)	Automatic positive reinforcement (sensory stimulation)
Negative (–)	Individually mediated negative reinforcement (direct avoidance or escape)	Socially mediated negative reinforcement (socially mediated avoidance or escape)	Automatic negative reinforcement (arousal reduction)

tion, or termination of an aversive social interaction, nonpreferred activity, or painful and unpleasant internal state, we classify those consequences as negative reinforcers. Third, behaviors maintained by automatic reinforcement produce pleasant sensory stimulation (automatic positive reinforcement) or alleviate unpleasant internal experiences (automatic negative reinforcement) without anyone or anything present. This point is critical for understanding the rationale for the “alone” condition in functional analyses; interfering behaviors exhibited when a student is alone point to an automatic reinforcement function. Behaviors such as stereotypy, self-injury, and certain habits (e.g., thumb sucking, nail biting, hair pulling, and binge eating) are those most likely to be maintained by automatic reinforcement.

Out of the Frying Pan . . . but into the Fun

In our experience conducting school-based FBAs, we often find that students' interfering behaviors are maintained by negative reinforcement contingencies: escape from “demands” or, as we like to reframe it, “avoidance or termination of instructional opportunities.” In many cases, these “escape-maintained” interfering behaviors also result in concurrent access to positive reinforcers such as preferred tangibles (e.g., “calming” sensory toys), activities (e.g., opportunities to engage in stereotypy or listen to music), and/or social interactions (e.g., one-on-one conversations with the teacher or a classmate).

Consider the case of an adolescent (ironically named “Myway”) who considers homework to be onerous and, in his words, “a royal pain in the ass . . . homework stinks.” When his parents instruct him to complete assigned homework, Myway typically displays verbal opposition and sometimes engages in property destruction. In response to these behaviors, Myway's parents reluctantly report that they often “give in” by negotiating alternative times to complete the assigned homework. The scenario usually ends with Myway getting “his way” by (1) avoiding the homework assignment and (2) engaging in alternative preferred activities such as watching television or playing video games. Thus, in this example, Myway's interfering behaviors are reinforced both by (1) socially mediated negative reinforcement and (2) individually mediated positive reinforcement. What a powerful cocktail—and one that needs to be addressed using a multicomponent intervention that targets both reinforcing contingencies!

ANTECEDENT VARIABLES

Antecedents are stimuli that occur *prior* to a behavior and make the behavior more probable. There are two types of antecedents that momentarily increase the likelihood of interfering behaviors: discriminative stimuli and motivating operations.

Discriminative Stimuli

Behaviors occur in the presence of specific stimuli. Eventually, the stimuli regularly associated with a behavior occurring and contacting reinforcement (we call this repeated association a learning history) serve as cues and increase the probability that the behavior will be performed (Kazdin, 2001). The stimuli that precede the occurrence of behavior and signal that reinforcement following the behavior is likely to occur are called *discriminative stimuli* (S^D s). In other words, S^D s signal the availability of reinforcement and thereby increase the probability of behavior. We are more likely to engage in behavior when the likelihood of reinforcement is maximized; the S^D is “telling us” that reinforcement is present and available. Unlike in respondent conditioning, when a stimulus directly elicits a response, in operant conditioning, an S^D *sets the occasion* for a subsequent response.

A behavior is said to be under *stimulus control* when responses emitted in the presence of a particular S^D produce reinforcement more often than responses that occur in the absence of the S^D . Or, stated in a slightly different way, a behavior that occurs at a higher rate in the presence of a given stimulus than it does in its absence is said to be under stimulus control (Cooper, Heron, & Heward, 2007). For example, we answer a phone when it rings, but not when it is silent, and we stop the car at intersections when the traffic signal is red, but not when it is green (Cooper et al., 2007). When a behavior is under stimulus control, it is highly probable (but not automatic) that the behavior will occur given the presence of a specific S^D .

Sometimes the S^D s are objects, and sometimes they are people. Consider a student who snatches food left out on the teacher’s desk. The food itself functions as an S^D for “food-snatching behavior.” Alternatively, consider a student who exhibits goofy behaviors that have been socially reinforced by eye contact and laughter from specific classmates. The student is more likely to engage in those behaviors when those classmates are present than when they are absent; therefore the classmates function as S^D s.

The Source of Reinforcement

When we conduct an FBA, we conceptualize an S^D as the *source of reinforcement*.

- When behaviors are reinforced by individually mediated reinforcement, specific items, locations, or situations that signal the availability of reinforcement operate as S^D s.
- When behaviors are reinforced by socially mediated reinforcement, the individuals who historically have delivered reinforcement operate as S^D s.

Examples of S^D s for Interfering Behaviors

The following case examples illustrate the effects of S^D s on interfering behaviors. *Hint: Words in bold and italics signal the S^D .*

- Sam is a sophomore in high school who has difficulty completing assignments at home (e.g., reading assigned chapters, writing papers, completing math computation assignments). Sam typically completes homework in her bedroom. Sam’s bedroom does not include a desk or chair, but does have a ***TV***. Several times a week, Sam goes to her bedroom with the best intentions of studying. However, after a few minutes of

studying, she turns on the TV and starts channel surfing. For Sam, the TV is an S^D for watching television. Removing the TV from her bedroom and adding a desk (an S^D for studying) may reduce the likelihood of watching TV and increase the likelihood of studying.

- A *paraprofessional* typically responds to a student's vocal outbursts by delivering reprimands and reminding the student about the importance of using an "indoor voice." The vocal outbursts continue to occur at a high rate, which suggests that social attention provided by the paraprofessional is reinforcing. These outbursts are most likely to occur when the paraprofessional is within earshot.
- A *parent* and adolescent recently had an argument. Initially, the adolescent requested to use her parent's *car*. The parent said no. This exchange escalated, and the adolescent started shouting and threatening to walk to her friend's house (3 miles away). The frustrated parent finally said, "Fine! Take the damn keys! But you better be home by 11:00 P.M. sharp!" Subsequently, the adolescent started arguments when the parent and the parent's car were both present.
- An adolescent opens the refrigerator and spies *18 bottles of cold beer*. He takes three bottles, knowing that his parents will not notice a few missing from a full refrigerator.
- A child is on a pediatric inpatient unit recovering from orthopedic surgery. Twice per day, an *occupational therapist (OT)* performs range-of-motion stretching with the child. The OT reports that the child "swears like a sailor" during therapy sessions. The psychologist conducts a behavior-analytic observation and confirms that swearing behavior does indeed occur within therapy sessions. The psychologist also notices a pattern. As the stretching intensifies, the child starts to complain about the pain. When the OT breaks out in song ("just a little bit longer . . . just a little bit more . . ."), the student exhibits swearing behavior, and the OT immediately terminates the stretching trial. The psychologist observes this pattern 12 times. In this case, the OT is an S^D signaling the availability of socially mediated negative reinforcement (i.e., escape from aversive therapy sessions . . . and perhaps from that falsetto singing).

Motivating Operations

Antecedents other than an S^D s also influence the occurrence of behavior (Iwata, Smith, & Michael, 2000). Laraway, Snyckerski, Michael, and Poling (2003) used the overarching term *motivating operation* (MO) to describe antecedent variables that *momentarily* alter the likelihood of behavior by changing the effectiveness (value) of consequences. MOs that evoke interfering behaviors generally involve either (1) the *deprivation of a preferred (reinforcing) stimulus* or (2) the *presentation of a nonpreferred (aversive) stimulus*. Deprivation of preferred social interactions, items, or activities tends to make students "want" those reinforcers even more, and the presentation of aversive social interactions and nonpreferred tasks tends to make students "want" to escape even more; accordingly, given these MOs, students are more likely to engage in interfering behaviors maintained by those momentarily valuable consequences. A single MO may influence multiple behaviors and consequences simultaneously (Mace, Gritter, Johnson, Malley, & Steege, 2006), but, within the context of an FBA, we are primarily concerned with identifying the MOs that "motivate" interfering behaviors.

Examples of MOs for Interfering Behaviors

The following case examples illustrate the effects of MOs on interfering behaviors. *Hint:* Words in **bold** and *italics* signal the MO.

- Remember Kent? Kent engages in wisecracking behaviors reinforced by social attention from classmates. His teacher decides to intervene by ***separating him from preferred classmates***. This intervention results in deprivation of preferred social interactions, which functions as an MO by temporarily increasing (1) the *value* of social attention as a reinforcer and (2) the *probability* of wisecracks and other forms of behavior that produce social attention (socially mediated positive reinforcement).

Bojangles: Now That's a Warning Stimulus!

During my (Steege) senior year of high school, I was enrolled in a history class with an outstanding and inspiring teacher, Mr. Bojangles. His knowledge of history and numerous stories about his travels around the world filled us with a sense of awe and opened our Midwestern minds to a vast array of challenges and possibilities. He used to say, "Anything is possible, dream big." I'll always remember him for his optimism and hope for the future. I'll also always remember him for one particular habit that over time became associated with the nastiest, smelliest, and foulest of odors that have ever exited a human body. Although he never offered any sort of explanation or medical diagnosis, he often punctuated class discussion with a penetrating release of gaseous fumes that would, well, gag a goat. But these "gas attacks" were not without warning. In fact, he had another odd habit of jingling coins in his pants pocket. Over time, we realized that the sound of jingling coins was soon followed by a choking concoction of deadly fumes. So here's the deal. In his attempt to cover the sound of flatulence, he would jingle the coins. But the jingle of coins only masked the *sound* of flatulence; it did not eliminate the offensive odor that was sure to follow. We quickly learned to associate the sound of jingling coins with the forthcoming odors. The jingling coins signaled a worsening aversive condition (a motivating operation). The jingling of coins acted as an early defense warning system that served to increase the value of escape from the classroom environment. Over time, students began to realize that the safest thing to do at the earliest sound of the jingling of coins was to exit the classroom. So, Mr. Bojangles would be leading an inspiring class discussion, we'd see him reach into his right pocket (the first warning stimulus), we'd hear the sound of jingling coins (the second warning stimulus), and then we'd immediately make excuses to leave the classroom. The mass exodus was comparable to scenes from the movie *Jaws* when panicked swimmers scrambled to the beach to escape shark-infested waters.

While this is clearly an exaggerated example, it does illustrate the *powerful* influence of antecedent variables. Now, let's turn our attention to some more relevant examples. Consider the following examples of warning stimuli MOs (highlighted in **bold** and *italics*).

- A student with a reading disability and performance anxiety engages in disruptive behavior when ***prompted to read orally***. The prompt is predictive of aversive events (e.g., making errors, experiencing unpleasant physiological arousal), establishes escape as a valuable reinforcer, and evokes escape-maintained interfering behaviors.
- When the classroom teacher says, "***OK, boys and girls, please get out your arithmetic books,***" Sandy often asks to go to the bathroom, her locker, the nurse's office, and so forth. This prompt is predictive of aversive events (i.e., dreaded arithmetic lessons), establishes escape as a valuable reinforcer, and evokes escape-maintained interfering behaviors.
- A student is engaged in free play (e.g., coloring). He sees the teacher ***organizing instructional materials*** and begins to make protest statements. The teacher's behavior is predictive of worsening (i.e., the end of coloring!), establishes avoidance as a valuable reinforcer, and evokes avoidance behaviors.

- Remember those mosquito bites and that “*itchy feeling*”? The itchy sensation is an MO. It is an aversive stimulus that temporarily increases (1) the *value* of relief (i.e., terminating the itching) and (2) the *probability* of scratching behavior that produces relief (automatic negative reinforcement).
- Al Gebra is a student with a history of academic delays in math computation. When *presented with a packet of math worksheets*, he immediately engages in verbal opposition and property destruction. Per the predetermined behavior plan, the teacher responds to these behaviors by directing Al Gebra to an isolation area where he is directed to think about his behavior and consider more appropriate ways to respond in the future. The math worksheets remain at his desk. The presentation of math worksheets (an aversive stimulus) functions as an MO by temporarily increasing (1) the *value* of task avoidance/termination and (2) the *probability* of interfering behaviors that result in task avoidance/termination (socially mediated negative reinforcement).
- Sebat is a student who receives one-on-one instructional support from a highly effective and preferred teacher. When another student, Jowayne, asks for assistance, the teacher *diverts her attention from Sebat* to help Jowayne. Sebat immediately exhibits self-biting behaviors. The teacher turns to Sebat and says, “Please stop. I don’t like it when you hurt yourself,” and Sebat stops. In this situation, the teacher diverting her attention is predictive of deprivation of attention. This is an MO that temporarily increases the *value* of teacher attention and *evokes* behaviors that historically have produced teacher attention (socially mediated positive reinforcement).
- Otitis is a student with a history of ear infections who is very sensitive to loud noises. The *fire alarm* is activated. Otitis immediately covers his ears with cupped hands and bolts from the classroom into the school courtyard. The fire alarm is an aversive stimulus (MO) that temporarily increases the *value* of escape and *evokes* escape-maintained behaviors (individually mediated negative reinforcement).

S^Ds and MOs

S^Ds and MOs share two important similarities: both (1) occur before behavior and (2) momentarily increase the probability of behavior. However, S^Ds signal that reinforcement is available and probable, whereas MOs alter the value of reinforcing consequences. The real-world distinction may be stated as follows: *MOs change how much people want something; S^Ds change their chance of getting it* (McGill, 1999).

MOs also directly influence S^Ds by increasing the effectiveness of S^Ds for occasioning behavior. To understand this influence, consider Brady, an adolescent with autism who had gained a considerable amount of weight over the past year. Due to health concerns, his team decided to place him on a diet to promote weight loss at the rate of 2 pounds per month. In addition to reducing his portions and providing him with low-fat meals, the team replaced preferred edible reinforcers such as chocolate chips, raisins, and chocolate-covered peanuts with healthier alternatives such as carrots, celery, and radishes. After 2 months on this plan, Brady had not lost any weight. In fact, he had gained 3 pounds. Even worse, Brady had a new interfering behavior in his repertoire: food stealing (e.g., taking food from others, raiding the refrigerator, and rummaging through the garbage).

I Need a Cup of Coffee

Imagine that you are on a road trip. You wake up early, decide to beat the rush-hour traffic, and hit the road. You are driving through an unfamiliar city. You really like your morning coffee, and you really prefer a certain brand of coffee (no gas-station or fast-food coffee for you!). For fun, let's say your preferred brand is "Star-Dunkin." You have not had any coffee for 26 hours, and that's basically a record for you. So, what do we know so far?

- Coffee (particularly Star-Dunkin coffee) functions as a reinforcer for you.
- You are seriously deprived of coffee (a motivating operation).

As you drive, you are on the lookout for a Star-Dunkin shop. You know they are popular in this region, so you are hypervigilant as you scour the passing malls and shopping centers where Star-Dunkin chains are typically located. So now, what else do we know?

- Star-Dunkin shops are S^Ds that signal the availability of reinforcement.

In sum, the MO (deprivation of coffee) makes coffee more valuable, purchasing coffee more probable, and Star-Dunkin shops more salient.

Lo and behold, you spy a Star-Dunkin shop! You pull over and purchase an 18-ounce mug of steaming-hot coffee. You get back into your car and continue on your trip. Within the next 5 minutes, you see two more Star-Dunkin shops. Do you pull over and get more coffee? Of course not. Your coffee fix has been met. The MO has been abolished, temporarily. The effect of coffee satiation is threefold:

- The value of coffee is momentarily reduced.
- The behavior of purchasing coffee is now less likely.
- Star-Dunkin shops as an S^D are now irrelevant. In fact, you may drive past the next three shops and barely notice them!

The moral of the story: You can't always get what you need . . . but sometimes you can get what you want—and it is *exactly* what you need at the moment.

Question: So, what's going on here?

Answer: Brady is experiencing food deprivation, which increases the value of restricted food items. Because of his increased *motivation* to obtain food, Brady is now highly observant of his environment and is constantly "on the lookout" for food. In fact, stimuli that had never previously occasioned food-seeking behavior are now functioning as "triggers" for food stealing. For example, when Brady sees the refrigerator, the food pantry, peers' lunch boxes, cafeteria trays, and the trash container in the kitchen, he is likely to attempt food-stealing behaviors. Thus, in this example, the deprivation of food (an MO) increases:

- The *value* of food as a positive reinforcer;
- The *probability* of food-stealing behavior; and
- The *effectiveness* of S^Ds in signaling the availability of food reinforcers and occasioning food-stealing behavior.

ANTECEDENTS AND CONSEQUENCES: A SUMMARY

MOs are stimuli or events that momentarily alter the value of consequences and the likelihood of behaviors maintained by those valuable consequences. The primary MOs that trigger interfering behaviors fall into two categories:

1. *Deprivation* of reinforcers (e.g., social attention, edibles, tangibles, and activities).
2. *Presentation* of aversive stimuli (e.g., unwanted social interactions, difficult tasks, and painful sensations).

S^D s signal the availability of reinforcement and therefore set the occasion for behavior. S^D s that trigger interfering behaviors may be conceptualized as *sources* of reinforcement, which may include:

- Actual reinforcing items or locations that signal the availability of reinforcers.
- People correlated with the delivery of reinforcement.

Reinforcing consequences are stimuli or events that follow and strengthen behaviors. Interfering behaviors may be maintained by one or more of the following forms of reinforcement:

- Individually mediated positive reinforcement (direct access).
- Individually mediated negative reinforcement (direct avoidance/escape).
- Socially mediated positive reinforcement (socially provided access).
- Socially mediated negative reinforcement (socially provided avoidance/escape).
- Automatic positive reinforcement (sensory stimulation).
- Automatic negative reinforcement (arousal reduction).

ADDITIONAL BEHAVIOR-ANALYTIC TERMS AND CONCEPTS

Response Classes (When It Rains . . . It Pours)

Sometimes behaviors with varying topographies have similar functional antecedents and consequences. (*This is just our cool way of saying that behaviors that look different may be very alike in that they share the same antecedents and consequences.*) In these cases, we say that these behaviors are members of the same response class. For example, behaviors such as hitting, kicking, slapping, and biting of others, while having different topographies, may be occasioned by the same discriminative stimuli, evoked by the same motivating operations, and reinforced by the same consequences. Oftentimes, these behaviors occur as a *response set*, which means that two or more of the behaviors in the response class occur within the context of a single behavioral episode. When it rains, it pours.

Response Class Hierarchies (First the Sky Darkens, Then There Is Thunder and Lightning Followed by a Light Rain . . . Then It Pours)

Sometimes, behaviors that are members of the same response class occur in a *predictable* sequence or chain. Often, the initial response in the chain is a low-intensity behavior that causes minimal disruption. When this behavior does not result in reinforcement, the student may escalate by exhibiting increasingly intense behaviors within the same response class. And when these behaviors are not reinforced, the student may display even more intense behavior, and so forth and so on, until a behavior results in the desired reinforcement. The terminal behavior in the chain tends to be one of high intensity and robust disruption.

What the *Func . . . tion?*

This is a true story from many years ago concerning my (Steege) two sons, Matt and Dan.

I came home from working in my position as a pediatric psychologist at the University of Iowa Hospital School inpatient unit, where we were conducting clinical research on the assessment and treatment of self-injury. When I walked into the house, I heard a loud slapping noise coming from the living room. I entered the living room to find Matt slapping himself, first on his arm, then on his leg, and then on his other arm. Of course, I did what any parent would do: I picked up the clipboard labeled “Matt’s Bizarre Behaviors” and began to collect data using a 6-second partial interval procedure to record occurrences of self-slapping behavior. Just kidding! I simply asked, “Matt, what are you doing?” He looked up and said, “Hey, Dad. I’m slapping myself.” I replied, “Hey, Matt. Why are you slapping yourself?” His next response was rather shocking: “Mom told me to.”

I immediately picked up another clipboard labeled “Lisa’s Unorthodox Behavioral Recommendations” and recorded my anecdotal observations and the results of the semistructured interview with Matt. Just kidding! Now, at this point it is probably important to note that Matt was about halfway through a bout of chicken pox. Those pox were at the really itchy stage, and Matt had been scratching at those pox to relieve the itchy feeling.

My wife, Lisa, who is now a special educator and a BCBA, recognized that the itchy feeling from the pox was an MO that established the effectiveness of automatic negative reinforcement (arousal reduction) and increased the probability of itch-reducing behaviors (i.e., scratching). Simply telling Matt, “Don’t scratch,” was only effective when she was present. So she decided to use differential reinforcement of incompatible behavior (DRI) by prompting Matt to *slap* the itchy pox as a replacement for scratching behavior. The slap had the same effect as scratching (arousal reduction—it reduced the itchy feeling), but without the harmful side effects. Hence, Matt’s self-slapping behavior. Problem solved. Time to unwind.

Just when I get this all figured out, little Dan comes running into the room after a long nap. He spied Matt on the couch and yelled, “Brubber! Look!” He then proceeded to vigorously slap his forehead with the open palm of his right hand. Matt, being a loving and caring older brother, found this to be hysterically funny and quickly said, “Dan, do it again.” And of course Dan did it again . . . and again . . . and again.

I then did what any parent would do. I picked up the clipboard labeled “Dan’s Goofy Behaviors” and began to record occurrences of self-slapping behaviors. This time I’m *not* kidding. All right, just kidding! Now, Dan was only 18 months old and had limited expressive communication skills. So asking him what was going on was out of the question. Also, Dan *did not have chicken pox* (yet!). So why was he slapping himself? His forehead didn’t itch. The slapping was not reducing an internal state of arousal. To figure out the function of Dan’s behavior, identify the following and then read on for the answers.

- What was the reinforcing consequence?
- What was the MO?
- What was the S^D?

So here’s the deal: I had two sons who engaged in responses that looked the same—self-slapping—but for very different reasons. Each son’s behaviors were evoked, occasioned, and maintained by different circumstances. In other words, they displayed similar behaviors with completely different functions. Matt engaged in self-slapping evoked by an itchy sensation and reinforced by a sense of relief. Dan engaged in self-slapping after a period of attention deprivation (MO), in the presence of his brother (S^D), to get his brother’s attention (socially mediated positive reinforcement).

The moral of the story: You can observe, record, and describe the form of a behavior, but to understand the function, you must conduct a functional assessment to identify behavior–environment relations.

Case Example: Response Class Hierarchies

Let's consider the case of Michelle, a student with an intellectual disability who really likes attention from her teachers. Michelle has a variety of behaviors in her repertoire for obtaining teacher attention (a reinforcing consequence). She exhibits a wide range of socially appropriate behaviors to access social attention. For example, depending on the situation, she may ask for help, say hello, or complete an assignment to show her teacher. Very infrequently, Michelle also engages in interfering behaviors such as clearing her throat loudly, yelling, "Hey, heyyy, heyyy," slapping the table, and picking at scabs on her hands. A comprehensive FBA indicated that these interfering behaviors all produced intermittent access to social attention. In other words, these "attention-seeking behaviors" formed a response class.

Due to the unexpected absence of two paraprofessionals and the unavailability of trained substitutes, the teacher-to-student ratio within Michelle's classroom abruptly shifted from 4:10 to 2:10. Michelle, who typically displays few, if any, interfering behaviors, is receiving much less attention from the classroom teachers than usual. (This reduction of attention functions as a motivating operation that increases the value of teacher attention as a reinforcer.) Michelle initially displays an appropriate behavior to gain teacher attention: politely asking for assistance. The teacher asks her to "wait her turn." Michelle then gets up from her desk, walks to the teacher, and politely asks for help again. The teacher asks her to "wait just a minute more." Several minutes pass without social interactions from the teachers, and Michelle clears her throat three times and says, "I need help, please." There is still no response from her teacher. She then loudly yells, "Heyyy, heyyy, heyyyyyyyy" and slaps the table six times. The teacher looks at Michelle and sternly says, "Michelle, you know better. Now behave." Michelle pauses and stares, like a famished lioness studying her prey, and then begins to pick at some scabs on her hand until they bleed. At this point, her teacher runs over and says, "Michelle, don't do that! You're hurting yourself and making a mess!" Finally . . . reinforcement! In our role as consultants, we see this sequence played out many times across all types of classrooms and grade levels, by students with varied exceptionalities who display all types of interfering behaviors.

The Matching Law

The matching law (Herrnstein, 1961, 1970) predicts that individuals will allocate responding across concurrent response options in direct proportion to the rate of reinforcement obtained for each response (i.e., relative rates of responding will *match* relative rates of reinforcement). In other words, when presented with behavioral choices, individuals typically spend more time engaging in behaviors that produce relatively more reinforcement. The matching law helps explain why students do what they do, and research shows that the matching law effectively predicts relative rates of appropriate and interfering behaviors in the classroom environment (Billington & DiTommaso, 2003; Martens, Halperin, Rummel, & Kilpatrick, 1990; Martens & Houk, 1989; Shriver & Kramer, 1997).

To illustrate the concept of the matching law, consider Bonita, a 12-year-old student with a behavioral disorder. Throughout her school day, Bonita faces a basic choice between engaging in appropriate behavior and engaging in interfering behavior. This simple choice situation is continuous, and she can choose to engage in either appropriate or interfering

behaviors at any given moment. Now, no matter how disruptive Bonita can be, she does not display interfering behavior for 100% of the school day. According to the matching law, the percentage of time that Bonita would be expected to engage in interfering behaviors is based primarily on the rate of reinforcement she obtains. When rates of reinforcement for appropriate behaviors are high and rates of reinforcement for interfering behaviors are low, Bonita is likely to spend relatively more time engaged in appropriate behaviors. In contrast, when rates of reinforcement for appropriate behaviors are low and rates of reinforcement for interfering behaviors are high, Bonita is likely to spend relatively more time engaged in interfering behaviors. Given that neither interfering behavior nor appropriate behavior results in reinforcement 100% of the time, engaging in both behaviors maximizes the amount of reinforcement she receives; therefore, sometimes Bonita may engage in appropriate behaviors and then switch to interfering behaviors, or vice versa. Effective intervention for Bonita, and indeed any student, thus requires *maximizing* the rate of reinforcement for appropriate behaviors and *minimizing* the rate of reinforcement for interfering behaviors.

Of course, human behavior is not as simple as Bonita's example implies, and multiple variables influence choice making. For example, in addition to the rate of reinforcement, the *value* of reinforcers and the *response effort* required to obtain reinforcement directly affect choice behavior (Cooper et al., 2007; Hoch, McComas, Johnson, Faranda, & Guenther, 2002; Mace & Roberts, 1993; Volkert, Lerman, & Vorndran, 2005). Reinforcement varies from being weak to robust, and the following *parameters of reinforcement* determine robustness:

- *Schedule.* The schedule of reinforcement roughly determines the rate of reinforcement, and students typically choose to engage in behaviors that produce higher rates of reinforcement.
- *Magnitude.* Magnitude refers to the quantity, duration, or intensity of the reinforcer. For example, an entire piece of pie is a larger-magnitude reinforcer than a single bite of pie. Students typically choose to engage in behaviors that produce higher magnitudes of reinforcement.
- *Quality.* The quality of reinforcement reflects preference. For example, if a student's most preferred beverage is root beer, then root beer is a higher-quality reinforcer than water. Students typically choose to engage in behaviors that produce higher-quality reinforcers.
- *Timing.* Timing refers to the immediacy of reinforcement. For example, there is a significant difference between earning a break now and earning a break in 2 hours. Students typically choose to engage in behaviors that produce more immediate reinforcers.
- *Response effort.* Response effort refers to the physical or cognitive energy required to perform a behavior. For example, the amount of effort required for a student with a reading disability and below-grade-level math computation skills to read and solve word problems ("It is February 1. Train A left the station in Boston traveling northeast to Portland, Maine, at an average speed of 40 mph. Train B left the station in New York traveling south to Miami at 90 mph. Which train do you want to be riding on?") may be far greater than the amount of effort required to say, "Nah. I'm not gonna do it." Students typically choose to engage in behaviors that produce reinforcement with

the least amount of effort. So, if an interfering behavior requires minimal effort, and the desired replacement behavior is extremely effortful, an intervention is unlikely to be effective.

It is important to consider each of these factors during the FBA process, as they have direct implications for designing effective treatment plans. To illustrate this point, let's revisit the case of Creed, the 8-year-old who exhibited interfering behaviors such as throwing materials, yelling at the teacher, climbing on desks, and running out of the room when presented with writing tasks. FBA results indicated the following:

- Writing assignments are difficult for Creed.
- Avoiding assignments that required writing was a valuable reinforcer.
- Interfering behavior always resulted in Creed being directed to the principal's office.
- Episodes of disruptive behavior ranged in duration from 20 seconds to 3 minutes before he was directed to the principal's office.
- Creed typically stayed alone in the principal's office without anything to do for 20–30 minutes before returning to the classroom.
- Creed was not required to complete missed writing assignments when he returned to the classroom.

Table 3.2 illustrates the robustness or “power” of the reinforcing consequences for Creed's interfering behaviors. This analysis suggests that the socially mediated negative reinforcement he receives for interfering behaviors is indeed powerful. In fact, if our goal was to increase and maintain a particular behavior, the parameters of reinforcement outlined in the case of Creed would likely accomplish just that! By conceptualizing Creed's behaviors in terms of the matching law, it becomes clear that we need to design an intervention that arranges very robust reinforcement for an appropriate replacement behavior (e.g., academic engagement, work completion, requesting a break, or asking for assistance). This may be accomplished by minimizing the value of reinforcement for interfering behaviors, while ensuring that the target replacement behavior is simple and produces consistent, immediate, and high-quality reinforcement.

TABLE 3.2. Parameters of Reinforcement Maintaining Creed's Behavior

<i>Schedule of reinforcement</i>	Dense: Fixed ratio 1 for each episode (i.e., each behavioral episode is followed by reinforcement).
<i>Magnitude of reinforcement</i>	High: 20–30 minutes of escape.
<i>Quality of reinforcement</i>	Moderate: Escape from a highly aversive task, but no access to highly preferred alternatives.
<i>Timing of reinforcement</i>	Variable: 20-second to 3-minute delay prior to reinforcement.
<i>Response effort</i>	Moderate: High physical effort, but less cognitive effort than required for writing.

In sum, an understanding of the matching law and an analysis of the parameters of reinforcement (both for target interfering and replacement behaviors) help us understand why students make certain choices and tell us what we're up against when we design interventions. Assessing the parameters of reinforcement allows us to answer the following questions:

- How robust is the reinforcing consequence that is maintaining the interfering behavior?
- How robust is the reinforcing consequence that is intended to maintain the logical replacement behavior?

For Creed, the most logical replacement behavior is the completion of writing assignments. However, assessment of the parameters of reinforcement provided for completing writing assignments showed the following: writing behavior resulted in intermittent reinforcement every 5–10 minutes (schedule), brief and relatively low-value reinforcement in the form of one to two statements like “nice job” (quality and magnitude), and fairly delayed reinforcement (timing). Additionally, the response effort for writing was high. The reinforcing contingencies for interfering behavior were far more powerful than the reinforcing contingencies for writing, and Creed's behavior is easily predicted by the matching law. No wonder Creed did what he did!

Reinforcement of Creed's Behaviors: Summary of Parameters

- Socially mediated negative reinforcement for interfering behaviors: **POWERFUL**
- Socially mediated positive reinforcement for writing behaviors: **TEENY-WEENY**

SUMMARY

Wow! Clearly the analysis of human behavior is not simple. Behavior is often the result of a complex interaction of multiple variables. This interaction is not necessarily linear, and behavior certainly is not static. Conducting an FBA requires consideration of all possible controlling antecedent and consequence variables and careful analysis of the reinforcing consequences for both interfering and appropriate behaviors. In the following chapters we discuss a variety of methods that may be used to tease out the variables that influence and maintain behavior.

To conclude this chapter, we offer resources for readers who are interested in learning more about the conceptual foundations of FBA and behavior-analytic interventions. There are numerous sources we could suggest, and we apologize if we have neglected seminal original works that would deepen your understanding of these principles and concepts. So, here's our “playlist.” Please add your own favorites.

1. Steege and Watson (2009)
 - Chapter 4: Everything You Always Wanted to Know about the Conceptual Foundations of Functional Behavioral Assessments . . . but Were Afraid to Ask: Or, Basic Principles of Functional Behavioral Assessment
2. Cooper, Heron, and Heward (2007)
 - Chapter 1: Definition and Characteristics of Applied Behavior Analysis
 - Chapter 2: Basic Concepts
 - Chapter 16: Motivating Operations
 - Chapter 17: Stimulus Control
 - Chapter 24: Functional Behavioral Assessment
3. Fisher, Piazza, and Roane (2011)
 - Chapter 7: Defining and Measuring Behavior
 - Chapter 8: Single-Case Experimental Designs
 - Chapter 10: Functional and Structural Approaches to Behavioral Assessment of Problem Behavior
 - Chapter 11: Indirect Behavioral Assessments: Interviews and Rating Scales
 - Chapter 12: Direct Observation
 - Chapter 19: Developing Function-Based Extinction Procedures
 - Chapter 20: Developing Function-Based Reinforcement Procedures
4. Mayer, Sulzer-Azaroff, and Wallace (2019)
 - Chapter 2: Designing Effective Strategies of Change: Essential Building Blocks
 - Chapter 5: Reinforcement: Fueling Behavior Change
 - Chapter 6: Increasing Behavior by Developing and Selecting Powerful Reinforcers
 - Chapter 10: Setting a Foundation for Positive Change: Identifying Participant's Functional Reinforcers
 - Chapter 11: Rapidly Attaining Positive Change: Implementing Reinforcement Effectively
 - Chapter 22: Maintaining Behavior: Ratio and Related Schedules of Reinforcement
 - Chapter 23: Maintaining Behavior by Arranging Time-Based Schedules of Reinforcement
5. NASP (*Best Practices in School Psychology*, 2014)
 - Chapter 18: Best Practices in Conducting Functional Behavioral Assessments
 - Chapter 20: Best Practices in Can't Do/Won't Do Academic Assessment
 - Chapter 27: Best Practices in Evaluating the Effectiveness of Interventions Using Single-Case Methods